

Q. 1: Prove Fermat's little theorem:

If p is a prime number and a is an integer such that $a \not\equiv 0 \pmod{p}$, then:

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof using modular arithmetic:

$$S = \{1, 2, 3, \dots, p-1\}$$

multiply each element in S by a , where $a \not\equiv 0 \pmod{p}$.

$$S' = \{a \cdot 1, a \cdot 2, a \cdot 3, \dots, a \cdot (p-1)\} \pmod{p}$$

$$\text{Thus, } a \cdot 1 \cdot a \cdot 2 \cdot \dots \cdot a \cdot (p-1) \equiv 1 \cdot 2 \cdot \dots \cdot (p-1) \pmod{p}$$

$$a^{p-1} \cdot (p-1)! \equiv (p-1)! \pmod{p}$$

$$\text{since } (p-1)! \not\equiv 0 \pmod{p}$$

$$a^{p-1} \equiv 1 \pmod{p}$$

$\therefore$ Fermat's little theorem is proved.

Given,

$$a = 7$$

$$p = 13$$

13 is prime and $7 \not\equiv 0 \pmod{13}$

$$7^{12} \equiv 1 \pmod{13}$$

$$7^{12} \pmod{13} = 1$$

1. Efficient modular Exponentiation.

$$c \equiv m^e \pmod{n} \text{ and } m \equiv c^d \pmod{n}$$

2. Key insights for Euler's theorem.

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

$$\phi(n) = (p-1)(q-1)$$

3. Primality testing.

$$a^{p-1} \not\equiv 1 \pmod{p}$$

Q.2: Euler's theorem statement.

compute $\phi(n)$ for $n \in \{35, 45, 100\}$.

if n is factored into distinct primes

$$n = p_1^{k_1} \cdot p_2^{k_2} \cdot \dots \cdot p_r^{k_r}$$

$$\phi(n) = n \cdot \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_r}\right)$$

$$\phi(35)$$

$$35 = 5 \cdot 7$$

$$\phi(35) = 35 \cdot \left(1 - \frac{1}{5}\right) \cdot \left(1 - \frac{1}{7}\right) = 35 \cdot \frac{4}{5} \cdot \frac{6}{7} = 24$$

$$\phi(45)$$

$$45 = 3^2 \cdot 5$$

$$\phi(45) = 45 \cdot \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{1}{5}\right) = 45 \cdot \frac{2}{3} \cdot \frac{4}{5} = 24$$

$$\phi(100)$$

$$100 = 2^2 \cdot 5^2$$

$$\phi(100) = 100 \cdot \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{5}\right) = 100 \cdot \frac{1}{2} \cdot \frac{4}{5} = 40$$

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

This is known as Euler's theorem.

Proof:

- $a \in \mathbb{Z}$
- $\gcd(a, n) = 1$
- $R = \{r_1, r_2, \dots, r_{\phi(n)}\}$

1. permutes the elements of R .

$$r_1, r_2, \dots, r_{\phi(n)} \equiv (ar_1)(ar_2) \cdots (ar_{\phi(n)}) \pmod{n}$$

$$r_1, r_2, \dots, r_{\phi(n)} \equiv a^{\phi(n)} \cdot r_1, r_2, \dots, r_{\phi(n)} \pmod{n}$$

$$N_1 = \frac{N}{n_1} = \frac{60}{2} = 30$$

$$N_2 = \frac{N}{n_2} = \frac{60}{3} = 20$$

$$N_3 = \frac{N}{n_3} = \frac{60}{4} = 15$$

$$N_4 = \frac{N}{n_4} = \frac{60}{5} = 12$$

Step 1:

$$x \equiv 1 \pmod{2}$$

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{4}$$

$$x \equiv 4 \pmod{5}$$

$$x \equiv 1 \pmod{2}$$

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{4}$$

$$x \equiv 4 \pmod{5}$$

Solve the system using the Chinese remainder theorem

$$N_i m_i \equiv 1 \pmod{n_i} \quad \frac{1}{n_i} \pmod{n_i}$$

Step-2

$$N_1 = 20 \pmod{n_1}$$

$$m_1 = 2:$$

$$20 \cdot 2 = 40 \equiv 1 \pmod{3}$$

$$m_1 = 2$$

$$N_2 = 15 \pmod{n_2}$$

$$15 m_2 \equiv 1 \pmod{4}$$

$$m_2 = 3:$$

$$15 \cdot 3 = 45 \equiv 1 \pmod{4}$$

$$m_2 = 3$$

$$N_3 = 12 \pmod{5}$$

$$12 m_3 \equiv 1 \pmod{5}$$

$$m_3 = 3:$$

$$12 \cdot 3 = 36 \equiv 1 \pmod{5}$$

$$m_3 = 3$$

Step-3:

$$x \equiv a_1 N_1 m_1 + a_2 N_2 m_2 + a_3 N_3 m_3 \pmod{N}$$

Given:

$$a_1 = 2, a_2 = 3, a_3 = 1$$

$$x \equiv 2 \cdot 20 \cdot 2 + 3 \cdot 15 \cdot 3 + 1 \cdot 12 \cdot 3 \pmod{60}$$

$$x \equiv 80 + 135 + 36 = 251 \pmod{60}$$

$$x \equiv 251 \pmod{60} \Rightarrow x \equiv 11 \pmod{60}$$

Q.4: Carmichael number:

$$a^{n-1} \equiv 1 \pmod{n}$$

Step-1

$$561 = 3 \times 187 = 3 \times 11 \times 17$$

561 prime factors: 3, 11, 17

Step-2:

prime factorization: 3, 11, 17

1. n is square-free (no repeated prime factors)

2. For every prime p dividing n , $p-1$ divides $n-1$

$$n-1$$

$$p-1 \text{ divide } 561-1 = 560$$

Step-3:

$$a^{560} \equiv 1 \pmod{561}$$

- $a=2$

$$2^{560} \pmod{561} = 1$$

- $a=10$

$$10^{560} \pmod{561} = 1$$

- $a=13$

$$13^{560} \pmod{561} = 1$$

Q.5:

Step-1

- $\phi(p) = p-1$

- $\{g^1, g^2, \dots, g^{p-1}\} \pmod{p} = \{1, 2, \dots, p-1\}$

- $p-1 \pmod{p}$

$$p=17,$$

$$\phi(17)=16$$

$$g^k \not\equiv 1 \pmod{17} \text{ for all } k < 16, \text{ but } g^{16} \equiv 1 \pmod{17}$$

Step-2:

- $g^{16/4} \not\equiv 1 \pmod{17}, 4 \text{ of } 16$

$$16 = 2^4 \Rightarrow 4 = 2 \Rightarrow g^8, g^4, g^2 \not\equiv 1 \pmod{17}$$

Step-3:

$$g = 3$$

$$3^1 \bmod 17 = 3$$

$$3^4 = (3^2)^2 = 81 \bmod 17 = 13$$

$$3^8 = (3^4)^2 = 13^2 = 169 \bmod 17 = 16$$

$$3^{16} = (3^8)^2 = 16^2 = 256 \bmod 17 = 1$$

powers = 1, only $3^{16} \equiv 1 \bmod 17$

Q-6:

$$3^x \equiv 13 \bmod 17$$

step-by-step solution:

x	$3^x \bmod 17$
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1	3
2	9
3	10
4	13

$$3^4 = 81 \bmod 17 = 13$$

$x = 4$ since, $3^4 \equiv 13 \bmod 17$

Q-7:

- p : a large prime
- g : a primitive root modulo p
- $g^a \bmod p$: party A's public key
- $g^b \bmod p$: party B's public key

steps of the protocol:

1. public parameters
2. private secrets
3. Exchange
4. shared secret computation

Discrete Logarithm

- g
- $g^a \bmod p$
- $g^b \bmod p$
- $g^a \equiv A \bmod p$

Security based on the DLP:

• $p > 2^{2048}$

• DLP in $\mathbb{Z}_p^*$

Q-8:

1. Substitution ciphering

Alphabet: $A \rightarrow D, B \rightarrow E, \dots, X \rightarrow A, Y \rightarrow B, Z \rightarrow C$

Plaintext: HELLOWORLD

Ciphertext: KHOORZRVOG

Result: "KHOORZRVOG"

Plaintext: HELLOWORLD

Digraphs: HE LL OW OR LD

• HE $\rightarrow H$ (row 2 col 2), E (row 3 col 1) $\rightarrow CF$

• LL $\rightarrow Lx \rightarrow L(4, 1) \times(5, 4) \rightarrow VS$

L $\rightarrow$ with filler $x \rightarrow VS$

• OW $\rightarrow O(1, 2) \omega(5, 3) \rightarrow NW$

• OR $\rightarrow O(1, 2), R(1, 5) \rightarrow NM$

• LD $\rightarrow L(4, 1), D(2, 5) \rightarrow TR$

Result: "CFVSUSNWMTR"

Q-9:

• Encryption function

$$E(x) = (ax + b) \bmod 26$$

• $a = 5$

• $b = 8$

plaintext: "Dept of ICT, MBSTU"

• x is the plaintext letter ($A \equiv 0, B \equiv 1, \dots, Z \equiv 25$)

• $\gcd(a, 26) = 1$ so $a = 5$ is valid:

part-a: Encrypt the plaintext:

Step-1

Dept of ICT MBSTU $\Rightarrow$ DEPTOFICTMBSTU

Step-2

$A \equiv 0, B \equiv 1, \dots, Z \equiv 25$

Part-B: Decryption

Step-1

$$D(y) = a^{-1} : (y - b) \bmod 26$$

• $a = 5$

• $a^{-1} \bmod 26 = 21, 5 \cdot 21 = 105 \equiv 1 \bmod 26$

Decrypted plaintext:

DEPTOFICTMBSTU

Q-10: Encryption process:

1. key and block structure
2. key derivation
3. substitution step.
4. permutation step.

Decryption process:

1. key derivation.
2. Inverse permutation.
3. Inverse substitution.

Basic cryptanalysis of sp-shift cipher

- simple and lightweight for educational use
- PRNG-driven substitution and permutation
- obscure direct letter-to-letter mapping.
- multi-round application increases confusion and diffusion.

improvements

- use a letter PRNG on a key schedule algorithm.
- Introduce inter-block chaining